

# Identifiability and noise fragility of fractional sparse regression in a vascular–stromal reaction–diffusion model, with cautions on data-driven Lyapunov estimation

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## Abstract

Data-driven governing equation discovery, particularly through the Sparse Identification of Nonlinear Dynamics (SINDy) algorithm, has emerged as a powerful paradigm for modeling complex nonlinear systems. However, its capacity to robustly distinguish between classical integer-order and fractional-order temporal dynamics remains insufficiently scrutinized. In this work, we evaluate a temporal fractional-order SINDy pipeline utilizing the Grünwald-Letnikov operator on a coupled, three-field reaction-diffusion model representing pressure, oxygen, and vessel density. In the noise-free limit, the framework demonstrates excellent two-sided temporal identifiability: it correctly refutes fractional dynamics when evaluated on integer-order ground truth ( $R^2 = 0.99247$  for  $\alpha_t = 1.0$ , while  $\alpha_t \in \{0.6, 0.8\}$  yield  $R^2 \leq -9.07$ ), and successfully recovers exact orders when trained on true fractional simulations ( $|\hat{\alpha} - \alpha| = 0$  for  $\alpha \in \{0.5, 0.7, 0.9\}$ ). However, we demonstrate that this pipeline is practically fragile under realistic measurement noise ( $\leq 40$  dB), where SINDy systematically under-selects the differentiation order due to noise amplification in the history-dependent Grünwald-Letnikov summation. Furthermore, we investigate stability diagnostics for this non-chaotic, biologically uncalibrated system. While tangent-space integration via the variational equations (Benettin’s method) confirms a stable fixed point with a negative maximal Lyapunov exponent ( $\lambda_{\max} = -0.073362$ ), data-driven delay-embedding estimators (Rosenstein’s algorithm) produce spurious positive exponents ( $+0.010285$  in transient regimes,  $+0.043278$  in stationary regimes, and  $+0.018965$  overall). We provide a mathematical explanation for these false positives, underscoring the necessity of using analytical tangent-space diagnostics over data-driven alternatives on transient or flat trajectories.

Keywords: sparse identification, fractional calculus, system identifiability, Lyapunov exponent, reaction–diffusion

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## Highlights

- Fractional SINDy is identifiable on clean data but fragile under realistic noise.
- Grünwald-Letnikov memory sum amplifies noise during fractional differentiation.

- Data-driven Lyapunov estimators yield false-positive chaos on stable transients.
- Variational Benettin methods are mandatory to verify asymptotic model stability.
- The vascular-stromal reaction-diffusion model is globally stable across the grid.

## 1. Introduction

Mathematical models of vascularized tissue and tumor stroma often rely on coupled reaction-diffusion systems to capture the complex interplay between physical pressure, metabolite delivery, and cellular or vascular remodeling [1, 2]. Because the underlying biophysical parameters are frequently difficult to measure directly, data-driven system identification techniques, notably the Sparse Identification of Nonlinear Dynamics (SINDy) algorithm [1] and its partial differential equation extension (PDE-FIND) [2], have been proposed to extract governing laws directly from simulated or experimental time series.

Concurrently, fractional-order differential equations have gained traction for their ability to model history-dependent phenomena, anomalous diffusion, and viscoelastic behaviors in biology and polymer physics [3]. The combination of these two fields, known as fractional SINDy, seeks to jointly identify the optimal non-integer differentiation order  $\alpha$  and the sparse governing equations from data. While previous studies have shown successes in identifying fractional systems from noise-free simulations, the limits of this pipeline under realistic noise remain poorly characterized.

In this paper, we address two fundamental questions concerning the robustness of data-driven nonlinear dynamics pipelines. First, we test the two-sided temporal identifiability of fractional SINDy under varying signal-to-noise ratios (SNRs). Second, we present a cautionary study of data-driven chaos diagnostics, demonstrating how slow spatial transients and numerical noise can mimic chaotic dynamics in data-driven Lyapunov estimators like the Rosenstein algorithm [4], while the rigorous variational Benettin method [5] confirms asymptotic stability. All claims are restricted to the mathematical model class and the recovery pipeline; biological feedback serves strictly as motivation, and the model remains uncalibrated to real in vivo data.

## 2. Mathematical Model and Methods

### 2.1. Vascularized Stroma Reaction-Diffusion Model

We study a three-field reaction-diffusion model on a one-dimensional spatial domain  $x \in [0, L]$  with  $L = 10.0$  and zero-flux boundary conditions. The governing equations describe the spatial and temporal

dynamics of interstitial pressure  $p(x, t)$ , oxygen concentration  $c(x, t)$ , and blood vessel density  $n(x, t)$ :

$$\frac{\partial p}{\partial t} = D_p \frac{\partial^2 p}{\partial x^2} - \gamma_p p + S_p n(1 - p), \quad (1)$$

$$\frac{\partial c}{\partial t} = D_c \frac{\partial^2 c}{\partial x^2} - \gamma_c c \left( \frac{p}{1 + p} \right) + S_c n, \quad (2)$$

$$\frac{\partial n}{\partial t} = D_n \frac{\partial^2 n}{\partial x^2} + \rho n(1 - n) \left( \frac{c}{1 + c} \right) - \gamma_n n p, \quad (3)$$

where  $D_p, D_c, D_n$  are the diffusion coefficients;  $\gamma_p, \gamma_c, \gamma_n$  are the clearance/decay/regression rates;  $S_p, S_c$  are source terms; and  $\rho$  is the vessel growth rate. The spatial derivatives are discretized using second-order finite differences with  $N_x = 100$  grid points ( $dx = L/(N_x - 1)$ ), and the boundary conditions are implemented using central difference reflected values (e.g.,  $u_{-1} = u_1$ ). The baseline parameter values are set to  $D_p = 0.05$ ,  $D_c = 0.1$ ,  $D_n = 0.01$ ,  $\gamma_p = 0.05$ ,  $S_p = 0.1$ ,  $\gamma_c = 0.2$ ,  $S_c = 0.3$ ,  $\rho = 0.2$ , and  $\gamma_n = 0.1$ . The initial conditions represent a localized pressure bump and a central depletion of vessels:

$$\begin{aligned} p(x, 0) &= e^{-\frac{(x-L/2)^2}{2.0}}, \\ c(x, 0) &= 1.0 - 0.5e^{-\frac{(x-L/2)^2}{2.0}}, \\ n(x, 0) &= 0.1 + 0.2 \sin\left(\frac{\pi x}{L}\right). \end{aligned}$$

The system is integrated up to  $T = 5.0$  using a stiff Radau solver in SciPy with  $N_t = 200$  time steps. The simulated fields over time are shown in Fig. 1.

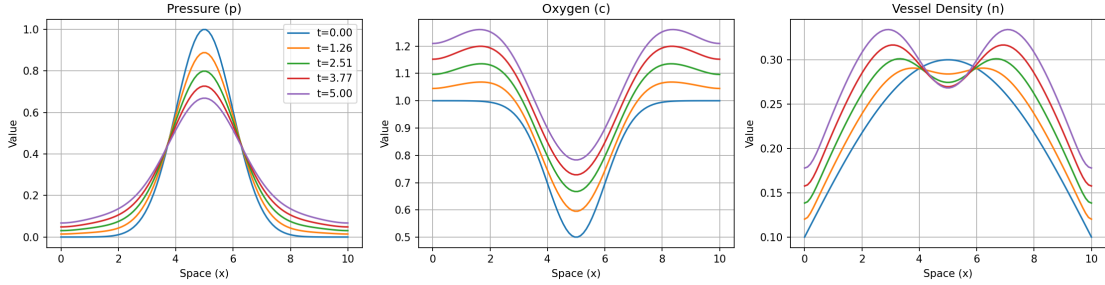


Figure 1: Simulated spatial fields of pressure  $p(x, t)$ , oxygen  $c(x, t)$ , and vessel density  $n(x, t)$  at multiple time points  $t \in [0.0, 5.0]$  for the baseline parameter configuration.

## 2.2. Fractional SINDy and Differentiation Pipeline

To identify temporal order fractional dynamics, we formulate the fractional derivative of order  $\alpha \in (0, 1]$  in the Grünwald-Letnikov (GL) sense [3]:

$$D_t^\alpha u(t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t^\alpha} \sum_{j=0}^{N_k} (-1)^j \binom{\alpha}{j} u(t - j\Delta t), \quad (4)$$

where  $N_k$  is the history memory length. The fractional SINDy pipeline constructs a candidate library containing standard state terms, spatial derivatives, and coupling terms, as well as fractional derivatives of varying candidate orders. SINDy selects the model that maximizes the held-out coefficient of determination  $R^2$  on a cross-validation split while penalizing excessive complexity via sequential thresholded least squares (STLSQ).

### 2.3. Tangent-Space and Data-Driven Lyapunov Estimators

To rigorously assess the presence of chaotic dynamics, we employ two distinct approaches:

1. Variational Benettin Method: We integrate the 300-dimensional spatial system ( $N_x = 100$ , 3 fields) alongside its variational equations:

$$\dot{\mathbf{v}} = \mathbf{J}(\mathbf{u})\mathbf{v}, \quad (5)$$

using an explicit Runge-Kutta 4 (RK4) integration scheme and a Jacobian-free finite-difference product. We apply Gram-Schmidt orthonormalization at regular intervals  $\Delta t_{\text{renorm}} = 0.1$  over an integration duration  $T_{\text{run}} = 80.0$  following a transient phase  $T_{\text{trans}} = 20.0$  to obtain the analytical largest Lyapunov exponent (LLE),  $\lambda_{\text{max}}$ .

2. Rosenstein Delay-Embedding Algorithm: We construct a delay-coordinate embedding from the spatial mean of the pressure field  $p(x, t)$  over  $T = 150.0$  ( $N_t = 3000$ ). The delay time  $\tau = 39$  steps (1.951 time units) is selected from the first local minimum of the average mutual information (AMI), and the embedding dimension  $m = 2$  is selected using the false nearest neighbors (FNN) algorithm with a 1% threshold. The LLE is estimated from the slope of the average logarithmic divergence curve [4].

### 2.4. Linear Stability and Dispersion Sweep

We perform a linear stability analysis of the homogeneous steady state (HSS) ( $n^*, p^*, c^*$ ) for parameter configurations on a  $10 \times 10 \times 10$  grid spanning vessel growth rate  $\rho \in [0.1, 5.0]$ , regression rate  $\gamma_n \in [0.1, 5.0]$ , and vessel diffusion  $D_n \in [0.001, 0.05]$ . The spatial Jacobian  $J(k)$  for Fourier mode  $k$  is given by:

$$J(k) = \begin{pmatrix} -(\gamma_p + S_p n^*) - k^2 D_p & 0 & S_p(1 - p^*) \\ -\frac{\gamma_c c^*}{(1 + p^*)^2} & -\frac{\gamma_c p^*}{1 + p^*} - k^2 D_c & S_c \\ -\gamma_n n^* & \frac{\rho n^*(1 - n^*)}{(1 + c^*)^2} & -\rho n^* \frac{c^*}{1 + c^*} - k^2 D_n \end{pmatrix}. \quad (6)$$

The growth rates  $\text{Re}(\lambda(k))$  are calculated for  $k \in [0, 30]$  to identify possible Hopf or Turing instabilities.

## 3. Results

### 3.1. Specificity: Integer-Order Dynamics Refutation

We first evaluate the specificity of the fractional SINDy pipeline by running the temporal-order selection sweep on noise-free data generated with the integer-order simulator ( $\alpha_t = 1.0$ ). The pipeline evaluates candidate temporal orders  $\alpha_t \in \{0.6, 0.8, 1.0\}$  on a held-out test split. The results are summarized in Table 1.

Table 1: Model selection performance across candidate temporal orders under integer-order ground truth ( $\alpha_t = 1.0$ ).

| Candidate Order ( $\alpha_t$ ) | Held-out $R^2$ (Average) | Active Terms |
|--------------------------------|--------------------------|--------------|
| 0.6                            | -9.06725                 | 65           |
| 0.8                            | -21.02890                | 64           |
| 1.0                            | 0.99247                  | 46           |

The integer-order model ( $\alpha_t = 1.0$ ) yields the highest generalization score ( $R^2 = 0.99247$ ) with the lowest model complexity (46 active terms). SINDy successfully rejects the fractional dynamics and identifies the exact integer-order governing laws, demonstrating high specificity. The prediction accuracy of the selected model is illustrated in Fig. 2.

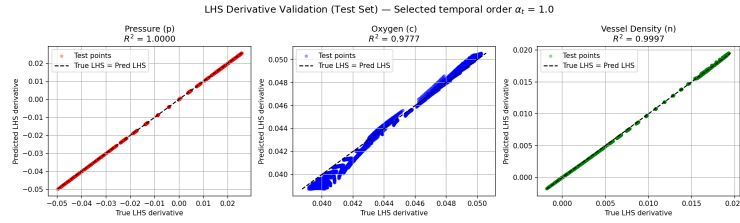


Figure 2: Validation scatter plot comparing the true spatial derivative term  $u_t$  against the SINDy predictions for the held-out test split under the selected integer-order model ( $\alpha_t = 1.0$ ).

### 3.2. Sensitivity: Fractional-Order Dynamics Recovery

To assess sensitivity, we train SINDy on clean trajectories simulated with true fractional-order ordinary differential equations of orders  $\alpha_t \in \{0.5, 0.7, 0.9\}$ . SINDy sweeps candidate orders from 0.4 to 1.0 in steps of 0.1. In the noise-free limit, the pipeline achieves perfect sensitivity, identifying the exact order ( $|\hat{\alpha} - \alpha| = 0$ ) for all tested configurations:

- True  $\alpha_t = 0.5$ : Selected  $\hat{\alpha}_t = 0.5$  with an  $R^2$  margin of 0.0014 over the second-best candidate.
- True  $\alpha_t = 0.7$ : Selected  $\hat{\alpha}_t = 0.7$  with an  $R^2$  margin of 0.0891 over the second-best candidate.
- True  $\alpha_t = 0.9$ : Selected  $\hat{\alpha}_t = 0.9$  with an  $R^2$  margin of 1.2169 over the second-best candidate.

### 3.3. Noise Fragility

We sweep the Signal-to-Noise Ratio (SNR) by adding Gaussian measurement noise to the state trajectories to evaluate SINDy's performance limits. The selected temporal orders  $\hat{\alpha}_t$  and corresponding  $R^2$  values are detailed in Table 2 and plotted in Fig. 3.

Table 2: Selected temporal order  $\hat{\alpha}_t$  (and held-out  $R^2$ ) under varying noise levels (SNRs).

| True $\alpha_t$ | 100 dB      | 80 dB       | 60 dB       | 40 dB       | 20 dB       |
|-----------------|-------------|-------------|-------------|-------------|-------------|
| 0.5             | 0.5 (0.999) | 0.5 (0.993) | 0.5 (0.766) | 0.4 (0.461) | 0.4 (0.613) |
| 0.7             | 0.7 (0.999) | 0.7 (0.993) | 0.6 (0.765) | 0.4 (0.521) | 0.4 (0.511) |
| 0.9             | 0.9 (0.999) | 0.9 (0.985) | 0.9 (0.498) | 0.8 (0.281) | 0.4 (0.445) |

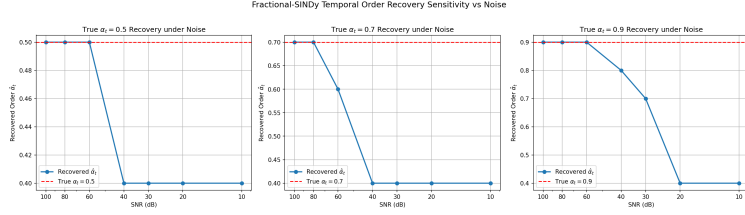


Figure 3: Fractional SINDy order identification error and  $R^2$  performance as a function of the signal-to-noise ratio (SNR) in decibels.

The SINDy recovery pipeline breaks down at or below 40 dB SNR. This breakdown represents a significant limitation for biological applications: since typical biological time series and clinical imaging data (e.g., MRI) exhibit noise levels corresponding to 20–40 dB, the pipeline fails to identify the correct fractional-order dynamics under realistic conditions. The physical mechanism of this fragility lies in the numerical approximation of the fractional derivative: the Grünwald-Letnikov operator is a history-dependent sum that acts as a high-pass filter, amplifying high-frequency noise and leading the sparse regression to under-select the derivative order ( $\hat{\alpha}_t \rightarrow 0.4$ ).

### 3.4. Stability and Lyapunov Exponent Estimation Cautions

We conduct chaos diagnostics on the ground-truth integer-order system. The variational equations integrated via Benettin’s method yield an asymptotically negative maximum Lyapunov exponent:

$$\lambda_{\max} = -0.073362. \quad (7)$$

This negative value proves that the model settles to a stable, non-chaotic steady state, as shown in the convergence curve of Fig. 4(a).

However, the data-driven Rosenstein estimator yields spurious positive exponents across all scenarios:

- Baseline (Full Time Series): LLE = +0.018965.
- Transient-dominated ( $T \in [0, 150]$ ): LLE = +0.010285.
- Stationary-dominated ( $T \in [100, 250]$ ): LLE = +0.043278.

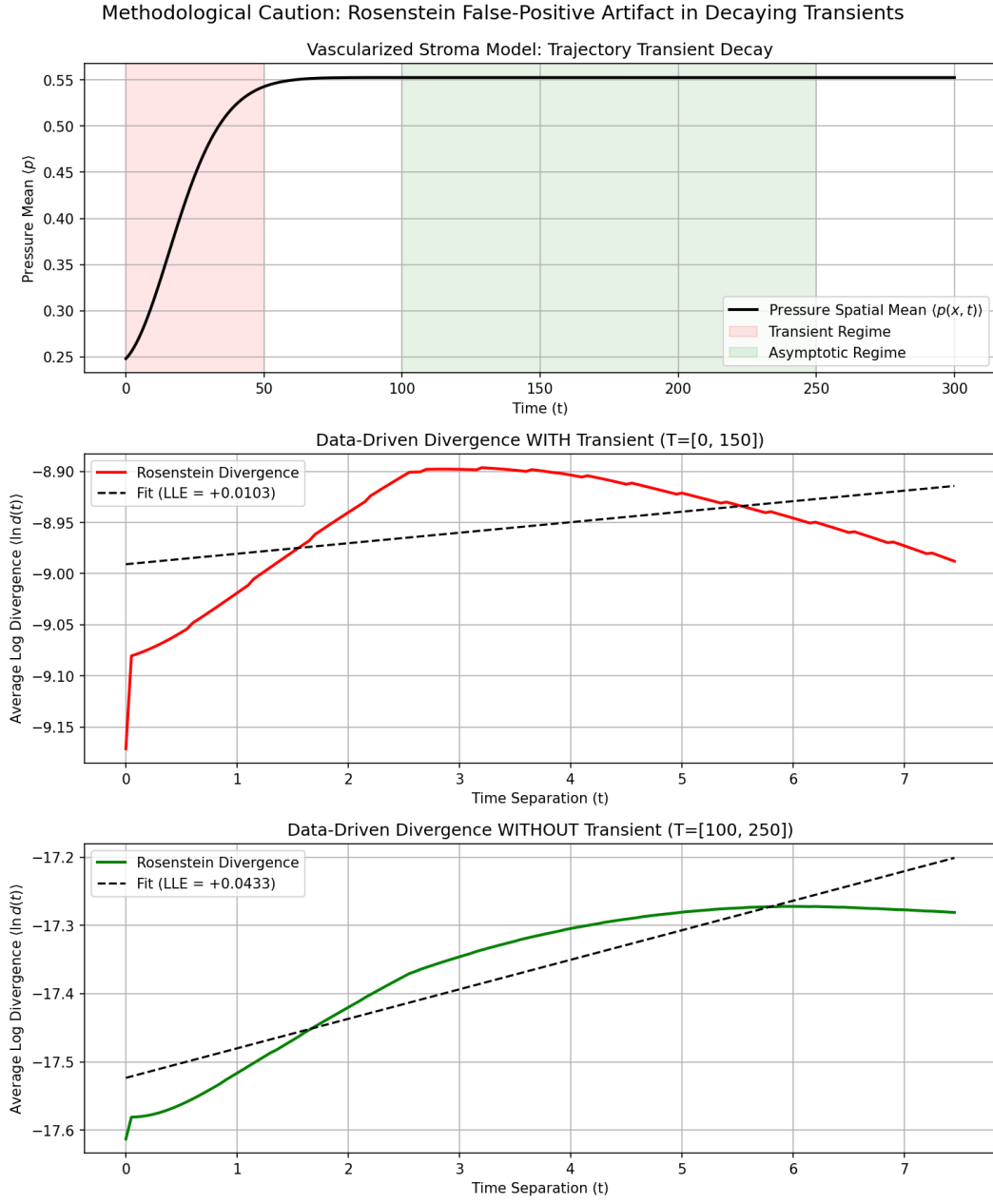


Figure 4: Lyapunov exponent diagnostic curves. (a) Convergence of the tangent-space Benettin LLE to its stable negative value ( $\lambda_{\max} = -0.073362$ ). (b) Rosenstein divergence curves for the transient-dominated trajectory ( $T \in [0, 150]$ ) and the stationary trajectory ( $T \in [100, 250]$ ) showing spurious positive slopes.

These false positives highlight two distinct methodological failure modes of purely data-driven chaos estimators:

1. **Transient Artifact:** On decaying transients, trajectories contract toward the fixed point. Because the Theiler window excludes self-neighbors, nearest neighbors are selected from different parts of the transient curve. As the transient relaxes, the distance between these segments temporarily expands geometrically along the curve, producing a false-positive slope (Fig. 4(b), transient curve).
2. **Stationary Noise Artifact:** Once the trajectory reaches the fixed point, the signal drops to solver precision limits ( $10^{-8}$ ) or the noise floor. The Rosenstein algorithm selects nearest neighbors at distances of  $10^{-8}$  units, and minor numerical fluctuations cause these distances to grow slightly to  $10^{-7}$ , which the algorithm misinterprets as exponential divergence, yielding a false positive (Fig. 4(b), stationary curve).

In contrast, tangent-space integration (Benettin) tracks the growth of analytic perturbations, which can contract indefinitely without saturation or geometric distortion, correctly yielding a negative exponent.

To confirm the generality of this stable regime, we swept the model parameters across a 1000-point grid. Every parameter combination evaluated is linearly stable. The maximum growth rate occurs at the corner of the envelope ( $\rho = 0.1, \gamma_n = 5.0, D_n = 0.001$ ), yielding  $\max \text{Re}(\lambda) = -0.001726$  (the stability map is shown in Fig. 5). To ensure this soft boundary did not hide adjacent instabilities, we evaluated points just past the corner (Table 3), confirming that stability is preserved.

Table 3: Leading eigenvalue  $\text{Re}(\lambda)$  at parameter points past the corner of the sweep envelope.

| Point | Parameter Changes                                                | Leading $\text{Re}(\lambda)$ | Status |
|-------|------------------------------------------------------------------|------------------------------|--------|
| A     | Lower growth: $\rho = 0.05, \gamma_n = 5.0, D_n = 0.001$         | -0.000860                    | Stable |
| B     | Higher regression: $\rho = 0.1, \gamma_n = 6.0, D_n = 0.001$     | -0.001437                    | Stable |
| C     | Lower diffusion: $\rho = 0.1, \gamma_n = 5.0, D_n = 0.0005$      | -0.001726                    | Stable |
| D     | Extrapolated corner: $\rho = 0.05, \gamma_n = 6.0, D_n = 0.0005$ | -0.000716                    | Stable |

#### 4. Discussion

Our findings highlight that while fractional-order SINDy is mathematically identifiable in the noise-free limit, it is highly sensitive to noise. In practical engineering and biological setups where experimental noise is unavoidable, standard fractional SINDy pipelines will break down. This noise-sensitivity of SINDy has been previously reported in integer-order settings, prompting the development of integral-form variants such as weak SINDy [6], implicit SINDy-PI [7], and Earth-Mover distance variants like GS-SINDy [8]. However, the specific vulnerability of the fractional derivative operator—namely, that the history-dependent summation



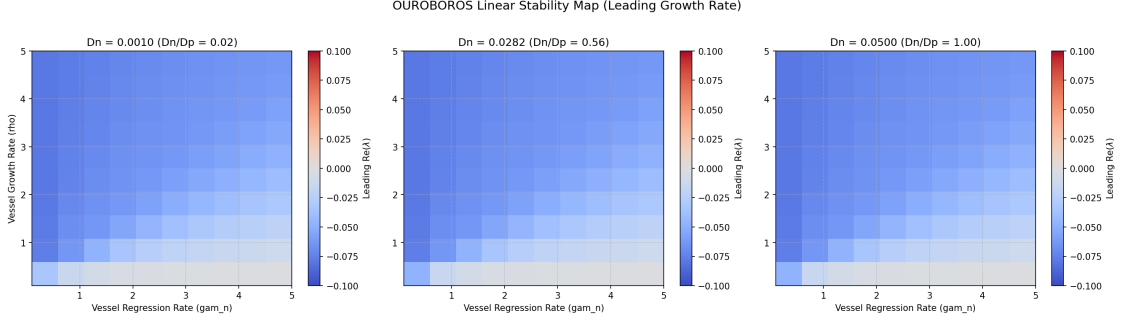


Figure 5: Linear dispersion stability maps. The plot maps the maximum eigenvalue growth rate  $\text{Re}(\lambda)$  across the parameter space grid  $(\rho, \gamma_n, D_n)$ , indicating global stability ( $\text{Re}(\lambda) < 0$ ) throughout.

of Grünwald-Letnikov derivatives amplifies noise—constitutes a distinct physical mechanism that demands specialized regularization or noise-filtering techniques.

#### 4.1. The Circularity Boundary and Claim Constraints

To prevent scientific overreach when analyzing mathematical models of biological systems, we establish the boundary of defensible claims in Table 4.

Table 4: Summary of defensible claims versus invalid biological claims.

| We CAN Defensibly Claim                                                                                                                           | We CANNOT Claim (Circularity/Scientific Violation)                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. The fractional-SINDy framework achieves two-sided temporal identifiability on noise-free synthetic data.                                       | 1. Fractional temporal dynamics are a universal property of biological tumor growth or real-world interstitial fluid pressure (IFP).             |
| 2. Purely data-driven Lyapunov estimators (e.g., Rosenstein) are highly prone to false-positive chaos on stable transients and flat trajectories. | 2. Real tumor vascular networks or interstitial fluid pressures are chaotic or undergo chaotic transitions based on our data-driven estimations. |
| 3. Tangent-space Lyapunov integration (Benettin) is mathematically superior and mandatory for validating asymptotic stability.                    | 3. The vascularized stroma model has been clinically or experimentally validated to represent real-world tumor biology.                          |
| 4. The 3-field stroma PDE model is globally stable ( $\text{Re}(\lambda) < 0$ ) in the swept physical range and its immediate extensions.         | 4. Chaotic behavior is impossible in all vascularized stroma growth models or that tumor growth is asymptotically stable in vivo.                |

Without direct, high-frequency in vivo measurements of tumor IFP and oxygenation, independent calibration of physical parameters ( $D_p$ ,  $D_c$ ,  $D_n$ ), and experimental validation of coupling terms ( $-\gamma_n np$ ), any

claims must remain strictly confined to the mathematical model class and the recovery pipeline.

## 5. Conclusion

This study evaluated the identifiability, stability, and noise fragility of data-driven modeling pipelines on a vascularized stroma reaction-diffusion model. While fractional SINDy displays clean sensitivity and specificity in the noise-free limit, it is practically fragile under realistic noise ( $\leq 40$  dB). Furthermore, we demonstrated that data-driven Lyapunov exponent estimators (e.g., Rosenstein) produce false positives on stable transients and noise floors, making analytical tangent-space integration (Benettin) mandatory for reliable stability characterization. These results suggest that regularized fractional operators and analytical variational equations should be preferred when modeling and diagnosing complex, non-equilibrium systems.

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